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A multi-robot monitoring system based on digital twin

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Abstract

Controlling multiple robots to work together without conflict is essential in the factories but it is difficult. The existing monitoring methods for multiple robots are not perfect because they are inconvenient to use and have poor real-time performance. To deal with these issues, this paper proposes a multi-robot monitoring system based on digital twin for simulation and collision detection of robot movements in the real environment. The overview framework of the proposed system and the related technologies will be introduced in this study. The designed system has the characteristics of friendly interface and convenient operation, which effectively improves the efficiency of condition monitoring and has practical significance for production monitoring.

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Keywords: Digital twin; multi-robot monitoring system; OPC UA

1. Introduction

Controlling the robots that work together without conflict is a difficult task in smart plants. Due to the large size of robot equipment, it is usually inconvenient to debug on site. Once multiple robots collide, it will bring huge economic loss. The existing monitoring methods are relatively simple, and are generally presented in the form of two-dimensional images, which cannot perform more detailed monitoring management. As an effective means of physical information fusion, digital twin technology can reflect the full life cycle process of the corresponding

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physical equipment in a three-dimensional virtual space, and has practical significance for realizing multi-robot monitoring.

Digital twin technology was first proposed by Professor Grieves [1] in his course at the University of Michigan in 2002. The significance of this theory lies in the realization of the feedback from the real physical space to the virtual digital space. It was mainly used to simulate and analyse the spacecraft in flight, monitor and predict the flight status of the spacecraft in the early years. In recent years, this technology has developed rapidly and has been widely used in many other fields such as ships, vehicles, power plants, complex electromechanical equipment, medical treatment, manufacturing workshops, and smart cities. Tao et al. proposed a digital-twin-based five-dimensional model[2], and verified it on wind power turbines[3]. Li et al. proposed an integrated development framework for the design and manufacture of complicated products based on digital twins[4]. In the field of monitoring, Zhao et al.[5] proposed a three-dimensional visualization monitoring method based on real-time information for the real-time visualization of digital twin factories. Using digital twin technology, Xiong et al.[6] designed a testing platform for cigarette factories. At present, some explorations have been carried out on monitoring based on digital twin, but no attempt has been made for monitoring the collaborative work of multiple robots.

This study proposes a multi-robot monitoring system based on digital twin. In the proposed system, robot-related data is obtained through OPC Unified Architecture (OPC UA) and Message Queue Telemetry Transmission (MQTT) protocols, and the movement of the robot in the real environment is simulated and collision detection based on the analysis of the data. The system helps to debug the collaborative control of multiple robots in a virtual scene, and thus can reduce the possibility of they collisions and conflicts.

The rest of this paper is organized as follows. First, we introduce the system framework of the proposed monitoring system in Section 2. Then, we introduce the main key technologies of the proposed system in Section 3. In Section 4, we show the visual interface of the system and the testing results of collision detection in the monitoring system. Finally, we summarize this study in Section 5.

2. System Framework

This study designs a multi-robot monitoring system based on digital twin. As shown in Fig.1, the system can be divided into five parts: the twin data, the real model, the virtual model, the platform service, and the connection.

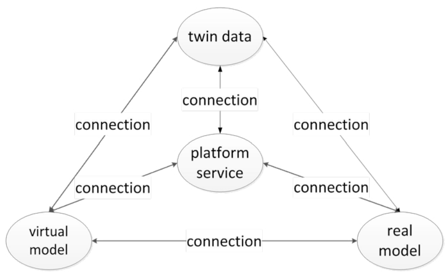

Fig. 1. The overall framework of the digital twin

Twin data is a necessary element to drive the operation of the system. The twin data mainly includes the environmental data where the device is located, the dynamic data generated by its own motion, and the constraint rules. Among them, the environmental data mainly includes equipment appearance data, factory appearance data, the equipment status, and the network status while dynamic data mainly includes sensor data, controller data, simulation data, and the results of evaluation, optimization, and monitoring.

The real model is the factory entity in the real scene and the foundation of the digital twin model. In this article, the real model refers to the production workshop, workshop equipment and workshop production process of computer numerical control (CNC) equipment.

The virtual model is to simulate the robot and its environment in a virtual three-dimensional scene, and the production activities in the real model. Through SolidWorks, 3dMax, Unity3D software for modeling, we can create a three-dimensional scene that is spatially consistent with the real environment, and drive the model to run through twin data.

Platform services are a collection of model display, data management and collision detection functions provided by the system. More specifically, it includes the simulation of robot movement, data statistics display of processing progress, display of equipment operation information, and alarms and data statistics analysis of multi-robot collisions. Virtual model, and the connection between platform services and the virtual model are realized.

The connection that is realized by the transmission of the data is another key to the interconnection of real model, virtual model, and platform service.

3. System key technology

The core of the system is to realize the mapping from the real model to the virtual model. In order to establish the mapping, reliable data is an essential factor to drive the system. Only under the drive of data can system platform services be provided and system monitoring functions can be realized. The key technology of the multi-robot monitoring system based on the digital twin can be summarized as the collection of twin data, the simulation of the robot movement in the virtual scene and the collision detection of the multi-robot movement.

3.1. Collection of digital twin data

Data collection is an important task of the digital twin system. There are two ways to collect twin data. One is to collect real-time data. The real-time data collected can simulate the real model and achieve the purpose of comprehensive monitoring. The other is to collect historical data in the cloud, and the function of system debugging can be realized by analyzing the historical data.

3.1.1. Data collection based on OPC UA

As the OPC UA architecture has the characteristics of high reliability, cross-platform, and access uniformity, this system adopts OPCUA-based data acquisition mode to collect the robot movement data which is dynamic. OPC UA can also realize real-time data collection through a publish-subscribe model, meeting the requirements of real-time mapping from real models to virtual models[7].

OPC UA can exchange data through the client/server model. The OPC UA server defines the address space and provides an accessible interface, and the OPC UA client implements data acquisition by calling the interface provided by the OPC UA server.

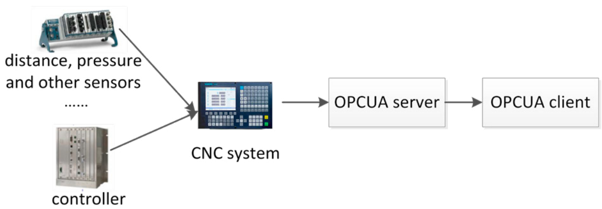

Fig. 2. Communication mode of OPC UA

The connection process between the OPC UA server and the client is that the OPC UA client first searches for the communicating server, and establishes a connection with the server after finding it. The server provides the device data required by the client through the address space. Then, the client accesses the address space of the server to read node data or subscribe to node data.

The data of the numerical control equipment is bound in the address space of the server through the node, and the client can access the address space of the server by subscription to obtain real-time data, which provides conditions for real-time mapping of the virtual model. The working principle of the protocol is mainly as follows: first, the client initializes a subscription service, and determines the node data that needs to be subscribed and the interval time for querying the data; and then the server sends subscription data to the client at equal intervals until the subscription ends. The whole process is shown in Fig.3.

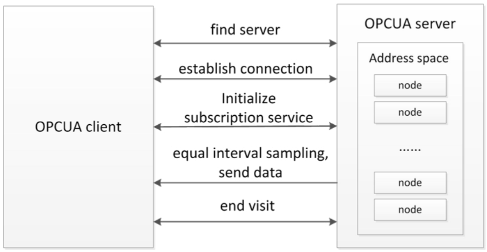

Fig. 3. OPC UA subscription model principle

3.1.2. Cloud historical data collection based on MQTT

In order to realize the debugging of the robot collaboration function, the system also supports the MQTT-based data collection method for collecting data from the cloud. MQTT that works on the TCP/IP protocol suite is a message protocol conforming to the publish/subscribe paradigm of the ISO standard (ISO/IEC PRF 20922) 8. Compared with the OPC UA protocol, it can achieve communication in remote devices with low hardware performance and poor network conditions.

The MQTT protocol usually has three identities: publisher, subscriber, and agent. Usually, the client plays the role of publisher and subscriber, and the server plays the role of proxy. The subscriber decides the message type (topic) to subscribe, and the agent returns the message content (payload) of the topic to the subscriber.

3.2. Motion simulation of virtual model based on robot kinematics

Using the twin data, the robot movement in the real model can be simulated. By the DH parameter method[9], the transformation matrix of each joint is multiplied in sequence to obtain the position of the robot end in the virtual scene relative to the base coordinates in the virtual scene. In this article, the six-degree-of-freedom manipulator is the research object, and the structure diagram is shown in Fig. 4.

Fig. 4. The structure of a six-degree-of-freedom manipulator

According to the parameters of the rods of the manipulator and the rules of the D-H coordinate system, the coordinate system of the manipulator is determined as shown in Fig.4, and the D-H parameters of the manipulator are listed in Table 1.

Table 1. D-H parameters.

Link number	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	$\theta_1(180^\circ)$
2	0	90°	d_2(200mm)	$\theta_2(0)$
3	a_3(410mm)	0	0	$\theta_3(90^\circ)$
4	a_4(410mm)	0	0	$\theta_4(0)$
5	0	90°	d_5(145mm)	$\theta_5(0)$
6	0	0	d_6(140mm)	$\theta_6(0)$

In Table 1, a_{i-1} refers to the length of the $(i-1)$th link, α_{i-1} refers to link twist, d_i refers to the link offset, and θ_i refers to the joint angle. The transformation matrix between adjacent joint coordinate systems:

$$
\begin{aligned}
{}_{i-1}^{i}T &= Rot(X, \alpha_{i-1})Trans(\alpha_{i-1}, 0, 0)Rot(Z, \theta_i)Trans(0, 0, d_i) \\
&= \begin{bmatrix} 0 & -\sin(\theta_i) & 0 & \alpha_{i-1} \\ \sin(\theta_i)\cos(\alpha_{i-1}) & \cos(\theta_i)\cos(\alpha_{i-1}) & -\sin(\alpha_{i-1}) & -\sin(\alpha_{i-1})d_i \\ \sin(\theta_i)\sin(\alpha_{i-1}) & \cos(\theta_i)\sin(\alpha_{i-1}) & \cos(\alpha_{i-1}) & \cos(\alpha_{i-1})d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
\end{aligned} \tag{1}
$$

The transformation homogeneous matrix from the base coordinate system $\{A_0\}$ to the end coordinate $\{A_6\}$ is as follows. Through this equation (1) and equation (2), the posture and the position of the end of the robotic arm relative to the base coordinate can be obtained.

$$
{}_6^0T = {}_1^0T\,{}_2^1T\,{}_3^2T\,{}_4^3T\,{}_5^4T\,{}_6^5T \tag{2}
$$

3.3. Collision detection of multi-robot movement

Collision detection is to simulate the reaction of objects in the real environment when they encounter obstacles and it is essential for the virtual operating system of industrial robots. If there is no collision detection in the system, the objects will penetrate or overlap. The collision detection algorithm used in this paper refers to [10]. The regular entity is quickly eliminated through bounding box detection. If there are overlapping bounding boxes, a further detection is performed to eliminate irrelevant points where collisions are impossible. Finally, an implicit function equation is used to solve the roots of the implicit function equation. The relationship between roots determines whether a collision occurs.

4. Result display and experimental simulation

The three-dimensional scene of multi-robot collision monitoring based on digital twin is shown in the Fig. 5, which can realize monitoring of the workshop robots. If the mobile device collides in the virtual model, the number of the machine that has the collision can be recorded and an alarm notification will be issued.

In the designed system, the robot can complete the same actions in the real model through the provided twin data. In addition, we can view the robot information (e.g., working status and production efficiency) and the detailed information of the lathes and machine centres.

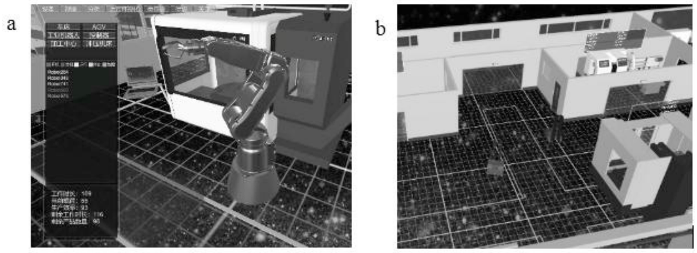

Fig. 5. (a) The working interface of the robot; (b) Collision scene

5. Summary

This article provides a design scheme for a multi-robot monitoring system based on digital twin. The system collects twin data based on OPC UA to realize the simulation monitoring of the motion of the six-degree-of-freedom robot, and test whether the collision will occur during the multi-robot collaboration. The designed system can simulate the real-world production and help to monitor the industrial robots, which enhances the production efficiency of the factories.

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